**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 101 · Binary Addition and Shifts

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Convert 00101101 binary to denary.

Q2. What is a character set?
A. Characters linked to codes
B. A list of fonts
C. A list of filenames
D. A collection of pixel colours

Q3. “A record must contain only one unnamed value.” Correct this. Explain why.

Q4. Valid ages are 13–18. Changing <18 to <=18 fixes a bug. Give three tests near 18 and their expected results. Why retest a value that worked before?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Carry — Value passed to the next binary column during addition.

Overflow — Result exceeds the largest value representable in fixed-width storage.

Logical left shift — Move bits left and insert zeros on right.

Logical right shift — Move bits right and insert zeros on left.

## New learning

Binary addition rules: 0+0=0; 0+1=1; 1+1=10, so write 0 and carry 1 into the next column.

Work right to left: Line up bits, add each column including any carry, then record the output.

Logical shifts: A logical left or right shift moves each bit by the stated number of places, introducing zeros at the vacated end.

Overflow and loss: With fixed eight-bit storage, a carry beyond bit 7 is lost (overflow); bits shifted out are discarded.

## Worked example

Add the eight-bit values 00010110 and 00001101, showing the carry where needed. [4 marks]

1. Line up eight positions

2. Add from the right: 0+1=1; next 1+0=1.

3. At the third bit, 1+1=0 carry 1; continue including the carry.

4. The eight-bit result is 00100011 (35 denary). [4]

# We Do · Practise with guidance

### Guided 1

Calculate 00101101 + 00000111 in binary. [4]

Worked method: 00101101 (45) + 00000111 (7) = 00110100 (52) [4; credit aligned working/carries].

### Guided 2

Perform a logical left shift of 00010101 by one place. [3]

Worked method: 00101010 [2]; all bits move one position left and 0 enters at the right [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## Guided 3

Perform a logical right shift of 10110000 by two places. [3]

Worked method: 00101100 [2]; two zeros enter on the left [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Change the multiplier to model a three-place logical left shift without overflow.

```
value = 9
result = value * 2
print(result)
```

write

### Question 1

Add 00011001 and 00000111 as eight-bit unsigned binary. [3] Source: Original OCR J277-style practice question.

### Question 2

Apply a logical left shift of two places to 00001011. [2] Source: Original OCR J277-style practice question.

### Question 3

Explain why adding 11111111 to 00000001 causes overflow in an eight-bit register. [3] Source: Original OCR J277-style practice question.

# Exit Ticket · Answer before checking

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

An algorithm only gives reliable results when its inputs are handled as specified.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. 45, because 32 + 8 + 4 + 1 = 45.

Q2. A — Characters linked to codes

Q3. Named fields about one item or person. Fields may have different data types.

Q4. 17 accepted; 18 accepted; 19 rejected. Retest a value that worked before to check for regression: a new fault caused by the fix.

### I Do / We Do pages · Guided practice

Q1. 00101101 (45) + 00000111 (7) = 00110100 (52) [4; credit aligned working/carries].

Q2. 00101010 [2]; all bits move one position left and 0 enters at the right [1].

Q3. 00101100 [2]; two zeros enter on the left [1].

### You Do pages · Code First

```
value = 9
result = value * 8
print(result)
```

Expected result: 72

### You Do pages · Independent questions

Q1. 00011001 (25) + 00000111 (7) = 00100000 (32) [3].

Q2. 00101100 [2].

Q3. The mathematical result is 256 [1], requiring a ninth bit [1]; the carry beyond eight bits cannot be stored [1].

### Exit Ticket · Model answer

True

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

## 1. Copy the whole original question:

## 2. Write a complete new answer in GREEN PEN. Show working if needed: